温州肯恩大学
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MGS 3101 W03 Foundations of Business Analytics

Multiple Linear Regression III

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Review

- What does a significant regression coefficient mean?
- Interpretation of a significant coefficient for:
  - A continuous IV
  - A categorical IV
  - A quadratic term
  - An interaction term
- Key takeaway
  - Start with the big picture, then consider details or context.
  - A quadratic term examines a nonlinear relationship; next, describe the form of nonlinearity.
  - An interaction term examines how the effect of one IV depends on another; next, explain how it changes.

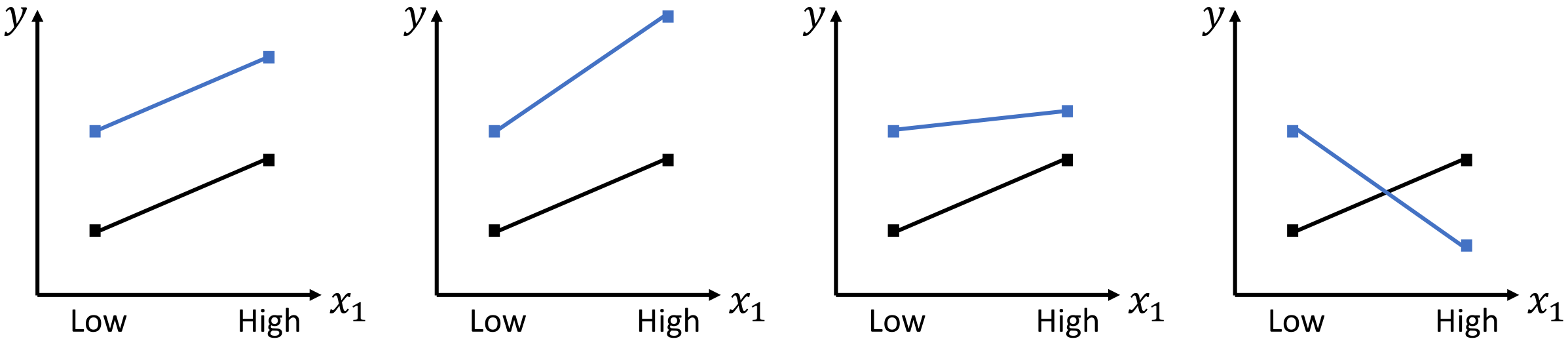
Recall *Food* × *Condiment*

Review

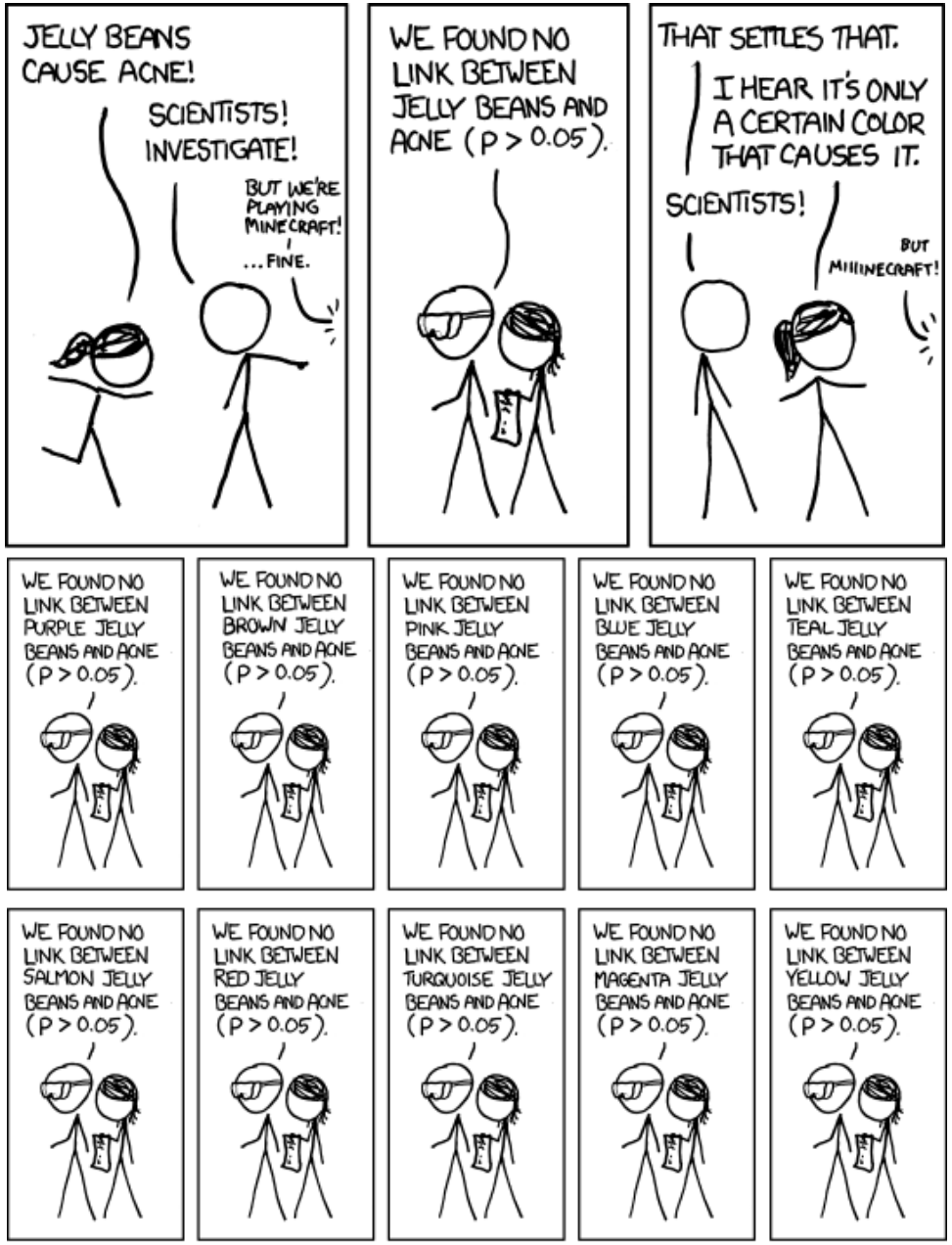
$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1 x_2$$

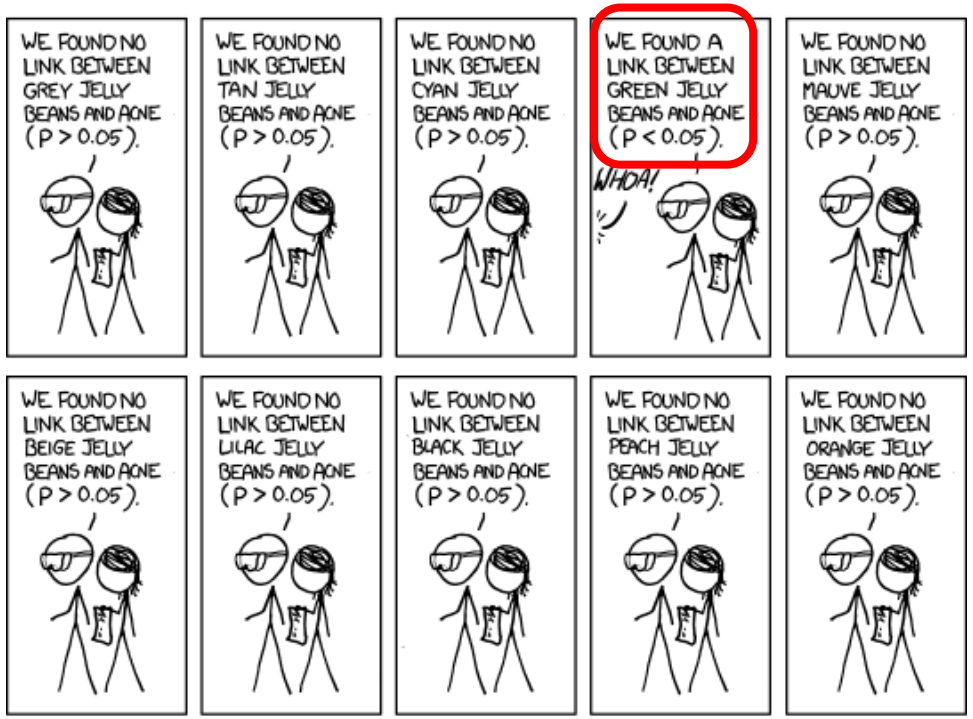
High x_2

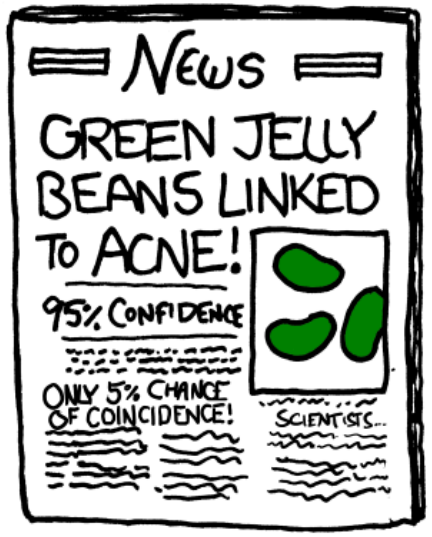
Low x_2



$\hat{\beta}_3^a$ $\hat{\beta}_3^b$ $\hat{\beta}_3^c$ $\hat{\beta}_3^d$







Do you believe it?

What does a significant regression coefficient mean?

Significant at 0.1/0.05/0.01 level

Significance

Table 4. Change in Number of Requested Third-Party Domains and Cookies

	Requests		Cookies	
	(1) EU firm	(2) Non-EU firm	(3) EU firm	(4) Non-EU firm
Post × *EU audience*	−0.0789***	−0.0464***	−0.1096***	−0.0722***
	(0.0026)	(0.0028)	(0.0036)	(0.0042)
Post × *non-EU audience*	−0.0504***	−0.0220***	−0.0674***	−0.0310***
	(0.0038)	(0.0011)	(0.0051)	(0.0015)
Observations	961,587	2,691,711	961,587	2,691,711
$\overline{R^2}$	0.8871	0.9054	0.8444	0.8641
Mean dependent variable	2.3706	2.4647	1.3476	1.4296

Notes. Dependent variable in columns (1) and (2) is the $\log(y+1)$ number of third-party domains that the website-host requests and in columns (3) and (4) the $\log(y+1)$ number of third-party domains that the website-host requests and respond with a cookie. *Post* indicates the period after May 25, 2018. *EU Audience* indicates whether a website host has a top-level domain that is specific to a country in the EU and/or whether the website's language is in a EU language and/or whether the website appears on any EU country's Alexa ranking. *Non-EU Audience* indicates the opposite. *EU firm* indicates that the website domain is associated with a firm that lists a headquarter in the EU on Crunchbase and/or a website where the Whois records indicate an owner with an address in the EU. Separate linear time trend for EU audience/non-EU audience before/after and website host fixed effects in all specifications. Standard errors in parentheses, clustered on the website-host-level.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Regulatory Spillovers and Data Governance: Evidence from the GDPR

Significance Increases with Sample Size

- Standard errors shrink as sample size increases
  - Smaller SE → larger t-value → smaller p-value → more “significant” results
- Noise averages out
  - More data reduces random variation, making estimates more precise and stable
- Hypothesis tests become more sensitive
  - With big n, even slight deviations from the null hypothesis are detected
- Large samples detect tiny effects
  - Even very small or practically unimportant coefficients can become statistically significant
- Key point
  - Statistical significance becomes easier with large samples, but significance ≠ practical importance
  - Focus on **effect size**, **confidence intervals**, and **business relevance**, not just p-value

Nonsignificance

- A coefficient is nonsignificant when its **p-value is large** (typically > 0.05).
- Interpretation: We do not have enough statistical evidence that this IV has a linear partial effect on the DV, after controlling for other IVs.
- Nonsignificance ≠ no real effect. The IV and DV may still have a linear relationship, but it may not be detectable in this sample.
- Common Reasons
  - Small sample size
  - Noisy or low-quality data
  - Multicollinearity
  - Model misspecification (missing nonlinear or interactions terms)

In reality, an IV and DV may have

- a linear relationship
- a nonlinear relationship
- or no relationship at all.

The Significant Problem of p-value

- p-hacking, data dredging / fishing / snooping
  - The manipulation of data analysis until it produces appear statistically significant results, compromising the truthfulness of the findings.
- Why p-values can mislead
  - A p-value measures how unlikely the sample data are **if the null hypothesis were true**, not the probability the hypothesis is true.
  - Large samples: even tiny and unimportant effects can become "statistically significant."
  - Small samples: real effects may fail to reach significance.
- Better practices
  - Focus on effect sizes, confidence intervals, and practical significance
  - Encourage replication and transparent reporting
  - Use p-values as **one piece of evidence**, not the sole decision rule

Big little lies: a compendium and simulation of p-hacking strategies

Common Forms of p-Hacking (Unethical, for Awareness Only)

Data Collection	Data Preprocessing	Hypothesis Testing	Reporting
• Stop data collection earlier than planned • Collect additional data without controlling for inflated error rate	• Transform distributions • Selectively exclude outliners	• Try multiple analytic models • Selectively control for covariates and moderators	• Selectively report outcomes and conditions • “Round off ” a p-value

The Myriad Forms of p-Hacking

How to Address Nonsignificance

Research Question:
Does employee job satisfaction improve job performance?

concept → measurement

not directly observable→ data

1. Revisit the business logic
   - Does the variable theoretically matter?
   - Is a relationship expected based on domain knowledge?
   - If theory strongly suggests relevance, consider keeping it even if nonsignificant.
   - If there is no meaningful business interpretation → candidate for removal.
2. Check data issues
   - Sample size too small? Small N → low power → coefficients often nonsignificant.
   - Measurement quality? Noisy or poorly measured variables weaken significance.
   - Outliers? Extreme values can inflate standard errors.
   - Missing values? Imputation or data loss can reduce reliability.

How to Address Nonsignificance

3. Investigate multicollinearity
   - Highly correlated predictors → inflated standard errors → nonsignificant coefficients
   - Detect using: Correlation matrix, VIF
   - If VIF is high: Remove or combine correlated variables, use domain judgment to select the more meaningful predictor, consider transformation or dimensionality reduction
4. Check model specification
   - Maybe the effect is nonlinear rather than linear.
   - Examples: Quadratic term, log transformation, interaction terms
   - If the true relationship is nonlinear, the linear coefficient may appear nonsignificant.

How to Address Nonsignificance

- Keep it when
  - Strong theory or business logic supports it inclusion.
  - It serves as a control variable needed to avoid omitted variable bias.
  - It is part of an interaction or nonlinear term (e.g., quadratic).
- Remove it when
  - It has no theoretical or practical relevance.
  - It contributes to multicollinearity or instability.
  - It adds unnecessary noise or complexity.
  - It does not improve prediction, especially in a purely predictive (not explanatory) model.

Variants of Regression Model (Overview)

- Linear & extended linear models
  - Multiple Linear Regression (MLR) – baseline linear relationship
  - Polynomial Regression – captures nonlinear patterns
  - Interaction Models – effect of one variable depends on another
  - Regularized Models (Ridge, Lasso) – handle multicollinearity & high-dimensional data
- Handling heteroskedasticity / correlated errors
  - Weighted Least Squares (WLS) – corrects heteroskedasticity
  - Generalized Least Squares (GLS) – handles correlated or non-constant variance errors
- Models for categorical outcomes
  - Logistic Regression (Binary / Multinomial / Ordered) – probabilities of categories
  - Probit Regression – similar to logistic with normal-based link

Variants of Regression Model (Overview)

- Count data models
  - Poisson Regression – event counts
  - Negative Binomial Regression – overdispersed counts
  - Zero-Inflated Models (ZIP / ZINB) – many zero outcomes
- Time series regression
  - ARIMA / SARIMA – autoregressive and moving-average models
  - Regression with ARMA Errors – regression + autocorrelated errors
  - State Space Models – dynamic systems with time-varying states
- Models for complex data structures
  - Panel Regression (FE, RE) – repeated observations over time and units
  - Survival (Cox) Regression – time-to-event outcomes
  - PLS Regression – many highly correlated predictors

Choosing the Proper Regression Model (Process)

1. Define the purpose of the analysis
   - Prediction → prioritize predictive accuracy, regularized models, ML-style regressions
   - Explanation / inference → interpretable linear or GLM models
   - Causal analysis → fixed effects, IV, DID, etc.
   - Classification → logistic / multinomial / probit
2. Identify type of DV
   - Continuous → Linear, Polynomial, Interaction, Regularized
   - Binary / Multinomial / Ordered → Logistic, Probit
   - Count → Poisson, Negative Binomial
   - Time-to-event → Cox (Survival)
   - Panel → FE / RE
   - Time series → ARIMA, regression with ARMA errors

Choosing the Proper Regression Model (Process)

3. Understand the data structure
   - Cross-sectional? time series? panel / longitudinal?
   - Repeated observations? correlated errors?
4. Check key assumptions and data issues
   - Linearity?
   - Heteroskedasticity vs homoskedasticity?
   - Autocorrelation?
   - Multicollinearity?
   - Zero inflation?
   - Overdispersion?
5. Evaluate model fit & diagnostics
   - Goodness of fit (R^2, AIC, BIC, deviance).
   - Residual checks.
   - Compare alternative models.

Logistic Regression – Motivation

Context:

We have a dataset of individual consumer purchases on an e-commerce platform, where we want to *predict a consumer's decision to purchase product Z* based upon other variables.

Customer	Last month's spending on platform	Marital status	Color of product shown to customer	Purchased?
Zhangsan	$400	Married	Black	Yes
Lisi	$366	Married	Black	Yes
Wangwu	$482	Single	White	No

Can we build a model to predict purchasing behavior?

Purchased ~ Spending + MarialStatus + Showncolor

How do we estimate the model when the DV is categorical (e.g. Yes vs. No)?

Logistic Regression – Motivation

- Categorical → Dummy: $y = 1$ is Yes, 0 if No.

Purchased	y
Yes	1
Yes	1
No	0

Any problem?

- Run the ordinal linear regression

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots$$

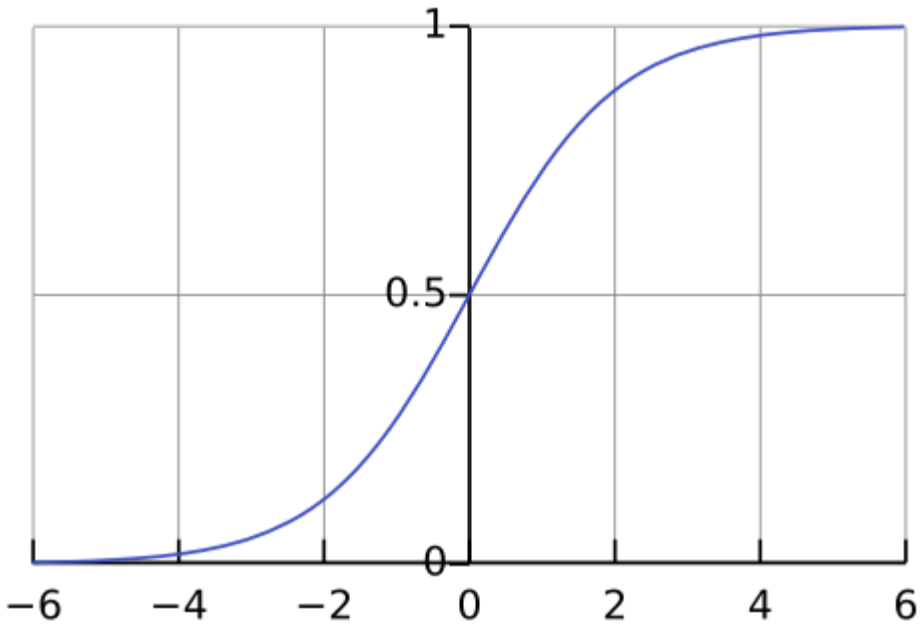
Logistic Regression – Motivation

Why linear regression does not fit when the DV is categorical?

- DV is not continuous
  - Ordinal categories do not have meaningful numeric distances (e.g., $1 \rightarrow 2 \neq 4 \rightarrow 5$).
- Predicted values may be invalid
  - Linear regression can output impossible values (e.g., 5.8, 0.3, −1).
- Violates homoskedasticity assumption
  - Categorical outcomes naturally produce non-constant variance.
- Errors are not normally distributed
  - Residuals take on only a few discrete values, violating OLS assumptions.

Logistic Regression – Motivation

Logistic function : $f(x) = \frac{1}{1+e^{-x}}$



Properties of logistic function

Logistic Regression

$$z = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k$$

$$p(y = 1|\boldsymbol{X}, \widehat{\boldsymbol{\beta}}) = \frac{1}{1 + e^{-z}}$$

linear combination $z \sim (-\infty, +\infty)$ $\xrightarrow[\text{function}]{\text{logistic}}$ Probability $[0,1]$

How to estimate this model, and how to interpret the regression coefficients?

Logistic Regression – MLE

- Definition
  - Maximum Likelihood Estimation (MLE) (最大似然估计) finds the parameter values that make the observed data **most likely**.
- How it works (Intuition)
  - Assume a probability model (e.g., Bernoulli, Normal, Poisson)
  - The model gives a probability for each observed data point
  - Multiply these probabilities together (i.i.d) → this is the likelihood
  - Choose parameter values that maximize this likelihood

Logistic Regression – MLE

Think:

MLE for linear regression estimation

Example

Suppose you observe 7 successes out of 10 trials. If you assume a **Bernoulli model** with probability p, the likelihood is:

$$L(p) = p^7(1-p)^3$$

MLE chooses the value of p that makes this observed pattern (7 successes, 3 failures) **most likely**.

$$\max_p L(p) = p^7(1-p)^3$$

The answer is:

$$\hat{p} = 0.7$$

Logistic Regression – MLE

Think:

Why don't we minimize the sum of square errors as in OLS?

- Each trial (purchase or not) follows a Bernoulli distribution
  - $y = 1$ (purchase) with a probability p
  - The probability p depends on customer characteristic $\boldsymbol{X}$ and parameters $\boldsymbol{\beta}$

$$z = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k$$

$$p(y = 1 | \boldsymbol{X}, \widehat{\boldsymbol{\beta}}) = \frac{1}{1 + e^{-z}}$$

- Goal: Choose $\boldsymbol{\beta}$ that maximizes the likelihood of observing the dataset.

$$\max_{\boldsymbol{\beta}} L(\beta) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1 - y_i}$$

Understand the loss function:

- If $y_i = 1$, we want p_i to be as close to 1 as possible.

$$\min_{\boldsymbol{\beta}} \log L(\beta) = -\sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Cost function 损失函数

Logistic Regression – Interpretation

$$p(y = 1|\boldsymbol{X}, \widehat{\boldsymbol{\beta}}) = \frac{1}{1 + e^{-(\widehat{\beta}_0 + \widehat{\beta}_1 x_1 + \widehat{\beta}_2 x_2 + \cdots + \widehat{\beta}_k x_k)}}$$

$$logit(p) \equiv log\left(\frac{p}{1-p}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k$$

- p is the probability of purchase ($y = 1$)
- p is nonlinear in $\boldsymbol{X}$, but log-odds are linear in $\boldsymbol{X}$.
- Interpretation
  - A 1-unit increase in x_i increases the log-odds by $\hat{\beta}_i$.
  - A 1-unit increase in x_i increases the odds by $e^{\widehat{\beta}_i}$.

odds/odds ratio: $\frac{p}{1-p}$

log-odds: $log\left(\frac{p}{1-p}\right)$

Logistic Regression – Interpretation

$$logit(p) \equiv log\left(\frac{p}{1-p}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

Coefficients:

	Estimate	Std. Error	z-value	Pr(>|z|)
(Intercept)	-4.005	2.432	-1.647	0.0996
x_1	0.395	0.165	2.393	0.017
x_2	-0.128	0.306	-0.418	0.676

- $\hat{\beta}_0$: **Expected** log-odds when x_1 and x_2 are both zero.
- $\hat{\beta}_1$: **Expected** increase in log-odds of event per unit-increase of x_1, holding x_2 constant.
- $\hat{\beta}_2$: **Expected** increase in log-odds of event per unit-increase of x_2, holding x_1 constant.

Logistic Regression – Interpretation

$$logit(p) \equiv log\left(\frac{p}{1-p}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

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x_2	-0.128	0.306	-0.418	0.676

- 0.395: Every unit-increase of x_1, holding x_2 constant, increases the log-odds by 0.395, or it *multiples* the odds by exp(0.395)=1.48, or the odds increase by 48%.
- For example, if x_1 is "Number of A-list celebrities endorsing your product", then getting one additional celebrity endorsement would, in expectation, increase each customer's odds of purchasing your product by 48%. (*not increasing probability, but odds*)

Logistic Regression – Summary

Think:

- What if we have more than 2 categories?
- Why model $P(Y = 1)$ instead of $P(Y = 0)$?

- What it does
  - Models the probability that $Y = 1$ (event occurs)
  - Uses the logistic function to keep predicted values between 0 and 1
  - Linear in log-odds, nonlinear in **probabilities**
  - A classification method, not true "regression"
- When to use it
  - DV is **binary** (e.g., buy/not buy, default/no default, churn/stay)
  - To predict event likelihood for decision-making (marketing, credit scoring, medical diagnosis, fraud detection)
  - To understand how predictors affect the log-odds of the event

Generalized linear model (GLM)

- Generalized linear model (GLM) (广义线性模型): A flexible generalization of ordinary linear regression,
  - by allowing the linear model to be related to the response variable via a ***link function***
  - by allowing the magnitude of the ***variance*** of each measurement to be a function of its predicted value.

Linear		Nonlinear
DV ~ Gaussian	$\xrightarrow[\text{Function}]{\text{Link}}$	DV ~ Exponential family of distribution
Homoskedasticity		Heteroskedasticity

Exponential Family of Distributions: A broad class of distributions that includes: Gaussian, Bernoulli, Binomial, Poisson, Beta, Dirichlet, Gamma, and many others.

From Regression to Causality

- You now know how to
  - Fit multiple linear regression
  - Interpret coefficients
  - Handle interactions, nonlinear terms, categorical IVs and DV
  - Evaluate significance and model fit
- But an important question remains
  - Does a significant coefficient imply a causal effect?

  → Not necessarily. Regression captures association, not causality.
- This leads us to the distinction between correlation vs. causality.

Correlation vs. Causality in Regression

- Correlation (Association)
  - Variables move together
  - Regression coefficients reflect partial associations
  - Useful for prediction and explanation
- But regression alone cannot tell us
  - Does X cause Y?
  - Does Y cause X?
  - Is a third variable causing both?
- Key point
  - Regression ≠ causal inference unless assumptions and design support it.

The Danger of “Pseudo-Regression” 伪回归

- Pseudo-regression
  - Running a regression and interpreting coefficients as causal without checking assumptions or the data-generating process.
- Typical problems
  - Omitted variable bias
  - Reverse causality
  - Confounders not controlled
  - Wrong functional form
  - Bad measurements
  - Spurious correlations
- Result
  - Significant coefficients that look “impressive” → but do not represent real causal effects.

Pseudo-Regression

$$\text{Ice Cream Sales} = \hat{\beta}_0 + \hat{\beta}_1 \text{Drowning Incidents}$$

- Regression output
  - $\hat{\beta}_1 = 0.82$, p-value < 0.001, $R^2 = 0.45$
  - Significant positive coefficient!
- False conclusion
  - More drowning incidents cause more ice cream sales, or, eating ice cream cause drowning.
- Real reason: a confounder (temperature / season)
  - Hot weather → more swimming → more drownings
  - Hot weather → more people buy ice cream
  - Temperature affects both variables, If temperature is omitted, regression produces a **spurious correlation**.

Pseudo-Regression

$$\text{Sales} = \hat{\beta}_0 + \hat{\beta}_1 \text{Advertising Spending}$$

- Regression output
  - $\hat{\beta}_1 = 2.5$, p-value < 0.001, $R^2 = 0.60$
  - Significant positive coefficient!
- Misleading conclusion
  - Increasing advertising spending causes sales to increase by 2.5 units per dollar.
- But this *<u>may be false</u>*!
- **Alternative explanation, confounder**
  - Seasonality as a confounder, influence both advertising and sales
    - Companies increase advertising during holiday seasons
    - Consumers buy more during holiday seasons
  - Without controlling for seasonality, regression may falsely attribute the seasonal sales increase to advertising, even if advertising had little or no true causal impact.

When Can Regression Support Causality?

- To argue causality, we need:
  - Temporal ordering: X must occurs before Y
  - Control of confounders: through variables, fixed effects, matching IV, etc.
  - Correct model specification: functional form and measurement must be appropriate
  - Plausible theoretical mechanism: clear explanation of how X affects Y.
- Causality requires design and assumptions, not just running a regression.

- Causal inference tools
  - Randomized experiments (RCT)
  - Natural experiments
  - Difference-in-differences (DID+PSM)
  - Instrumental variables (IV)
  - Regression discontinuity (RDD)
  - Panel fixed effects

Proving causality is difficult, and using observational data makes it even more challenging — often impossible.

Summary

- Significance ≠ Importance
  - Large samples make statistical significance easy.
  - Always evaluate effect size, confidence intervals, and business relevance.
- p-values can mislead
  - Multiple testing and p-hacking create false “significant” results.
  - Use p-values as one piece of evidence, not the final decision rule.
- Handling nonsignificance
  - Check theory, data quality, multicollinearity, and model specification.
  - Keep variables that are theoretically essential; remove those that add noise.

Summary

- Choosing the right regression model
  - Match the model to the DV type, purpose (prediction vs. explanation), and data structure.
  - Options include: MLR, polynomial, interaction models, WLS/GLS, logistic, Poisson, panel, time series, etc.
- Logistic regression & MLE
  - Logistic models predict probability, not continuous outcomes.
  - Parameters are estimated via Maximum Likelihood, not OLS.
- Correlation vs. Causation
  - Regression captures associations, not causal effects.
  - Causal inference requires strong design (RCTs, IV, DID, RDD, FE, etc.).